

Zeros, Multiplicity, and Graph Behavior

Answer Key

Algebra 2

Quick Answers

- | | |
|--|--|
| 1. 2, 8 dollars | $(x - 2)(x + 2)(x - 3)(x + 3)$, (b) the zeros = |
| 2. (a) the zeros = $-3, 1, -$ | $-3, -2, 2, 3$ |
| 3. (a) factored form = $x(x - 4)(x - 5)$, (b) the | 7. (a) zeros of the model you wrote = $-1, 4$, (b) |
| zeros = $0, 4, 5$ | standard form = $x^3 - 2x^2 - 7x - 4$ |
| 4. (a) the zeros = $1, -4$, (b) the y-intercept = | 8. -2 |
| -16 | 9. (a) the zeros = $-1, 2$, (b) $f(1) = 2, -$ |
| 5. (a) $p(1) = -12, -$ | 10. (a) factored form = $x(x - 2)^2$, (b) the zeros |
| 6. (a) factored form = | = $0, 2, -$ |
-

What is verified

16 of 20 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 2, 5, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSA-APR.B.3 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–4 of 5 across 20 tagged checks.

- 1.** Break-even prices for $P(x) = (x - 2)(x - 8)$, profit in dollars, x the price per cup in dollars.

Solution: A product is zero exactly when one of its factors is zero. $x - 2 = 0$ gives $x = 2$; $x - 8 = 0$ gives $x = 8$. Both are sensible prices, so the stand breaks even at

$x = 2$ or $x = 8$ dollars

- 2.** $g(x) = (x + 3)^2(x - 1)$. (a) All real zeros. (b) The zero where the curve turns back, with a reason.

Solution (a): Set each factor to zero: $(x + 3)^2 = 0$ gives $x = -3$, and $x - 1 = 0$ gives $x = 1$.

$x = -3$ or $x = 1$

Solution (b): *Instructor-judged.* The zero is $x = -3$. The factor $(x + 3)$ appears to the second power, so that zero has multiplicity 2 — an even multiplicity. Near an even-multiplicity zero the factor $(x + 3)^2$ never changes sign, so g keeps the same sign on both

sides and the curve meets the axis and turns back instead of passing through. A full-credit answer names $x = -3$ and gives the even-multiplicity reason.

3. Tray volume $V(x) = x^3 - 9x^2 + 20x$ cubic centimetres. (a) Factor completely. (b) Values of x with zero volume.

Solution (a): Every term carries an x , so take the greatest common factor first:

$$V(x) = x(x^2 - 9x + 20).$$

Then $x^2 - 9x + 20$ needs two numbers with product 20 and sum -9 : those are -4 and -5 .

$$V(x) = x(x - 4)(x - 5)$$

Solution (b): Each factor gives one zero: $x = 0$, $x = 4$, $x = 5$.

$$x = 0, 4, 5$$

4. $f(x) = (x - 1)(x + 4)^2$. (a) All real zeros. (b) The y -intercept $f(0)$.

Solution (a): $x - 1 = 0$ gives $x = 1$; $(x + 4)^2 = 0$ gives $x = -4$ (multiplicity 2).

$$x = 1 \text{ or } x = -4$$

Solution (b): Substitute $x = 0$: $f(0) = (0 - 1)(0 + 4)^2 = (-1)(16)$.

$$f(0) = -16$$

5. $p(x) = -(x + 2)(x - 3)^2$. (a) $p(1)$. (b) End behavior, and behavior at $x = 3$.

Solution (a): $p(1) = -(1 + 2)(1 - 3)^2 = -(3)(4)$.

$$p(1) = -12$$

Solution (b): *Instructor-judged.* Multiplying out, the leading term is $-x^3$: odd degree with a negative leading coefficient. So the curve rises without bound far to the left and falls without bound far to the right. At $x = 3$ the factor $(x - 3)$ is squared, an even multiplicity, so the curve meets the axis there and turns back rather than crossing. A full-credit answer states both ends correctly and names even multiplicity as the reason for the turn.

6. $q(x) = x^4 - 13x^2 + 36$. (a) Factor completely. (b) All real zeros.

Solution (a): This is a quadratic in x^2 . Two numbers with product 36 and sum -13 are -4 and -9 :

$$q(x) = (x^2 - 4)(x^2 - 9).$$

Each bracket is a difference of two squares.

$$q(x) = (x - 2)(x + 2)(x - 3)(x + 3)$$

Solution (b): One zero per factor.

$$x = -3, -2, 2, 3$$

7. Build a degree-3 model that turns back at $x = -1$, crosses at $x = 4$, leading coefficient 1. (a) Factored form and zeros. (b) Standard form.

Solution (a): Turning back at $x = -1$ needs even multiplicity, and degree 3 with a single crossing at $x = 4$ leaves multiplicity 2 for $x = -1$:

$$M(x) = (x + 1)^2(x - 4).$$

Its zeros are

$$x = -1 \text{ or } x = 4$$

Solution (b): Expand $(x + 1)^2 = x^2 + 2x + 1$, then multiply by $x - 4$:

$$\begin{aligned}(x^2 + 2x + 1)(x - 4) &= x^3 + 2x^2 + x - 4x^2 - 8x - 4 \\ &= x^3 - 2x^2 - 7x - 4.\end{aligned}$$

$$M(x) = x^3 - 2x^2 - 7x - 4$$

8. $C(x) = a(x - 1)(x - 4)(x - 6)$ dollars, with $C(0) = 48$ dollars. Find a .

Solution: Substitute $x = 0$ and set the result equal to 48:

$$\begin{aligned}a(0 - 1)(0 - 4)(0 - 6) &= 48 \\ a(-1)(-4)(-6) &= 48 \\ -24a &= 48.\end{aligned}$$

Dividing by -24 gives

$$a = -2$$

9. $f(x) = (x + 1)(x - 2)^2$. (a) Zeros. (b) $f(1)$. (c) Sketch showing the behavior at each zero.

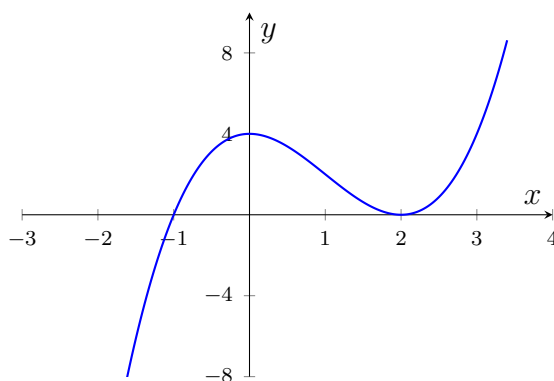
Solution (a): $x + 1 = 0$ gives $x = -1$ (multiplicity 1); $(x - 2)^2 = 0$ gives $x = 2$ (multiplicity 2).

$$x = -1 \text{ or } x = 2$$

Solution (b): $f(1) = (1 + 1)(1 - 2)^2 = (2)(1)$.

$$f(1) = 2$$

Solution (c): *Instructor-judged.* A correct sketch crosses the x -axis at $x = -1$, meets the axis at $x = 2$ and turns back without crossing, passes through $(1, 2)$, and — since the leading term is x^3 — falls far to the left and rises far to the right. The model curve is shown below.



10. Berm height $R(x) = x^3 - 4x^2 + 4x$ metres. (a) Factor completely. (b) Zeros. (c) Meaning of the repeated zero.

Solution (a): Take out the common x , then recognise a perfect square:

$$R(x) = x(x^2 - 4x + 4) = x(x - 2)^2.$$

$$R(x) = x(x - 2)^2$$

Solution (b): $x = 0$ from the first factor and $x = 2$ from the squared factor.

$$x = 0 \text{ or } x = 2$$

Solution (c): *Instructor-judged.* The zero $x = 2$ has multiplicity 2. Because that factor is squared it never changes sign, so the berm height comes down to the path at 2 metres along, touches it, and rises again — it does not pass below the path. At $x = 0$ the single factor means the height does cross the path level. A full-credit answer reads the even exponent as the reason for the turn.